

```
import pandas as pd
```

```
df = pd.read_csv("/content/Salary_dataset.csv")
```

```
df
```


	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0
5	5	3.0	56643.0
6	6	3.1	60151.0

```
df.head()
```

	Unnamed: 0	YearsExperience	Salary
8	8	3.3	64446.0
0	0	1.2	39344.0
10	10	4.0	63219.0
1	1	1.4	46206.0
2	2	1.6	37732.0
12	12	4.1	56958.0
3	3	2.1	43526.0
4	4	2.3	39892.0
14	14	4.6	61112.0

```
X = df.YearsExperience
Y= df.Salary
```

```
import numpy as np
```

```
X=np.array(X).reshape(-1,1)
Y=np.array(Y).reshape(-1,1)
```

```
X
```

22	22	8.0	101303.0
array([[	1.2],		
23	[1.4]23	8.3	113813.0
	[1.6],		
24	[2.1]24	8.8	109432.0
	[2.3],		
25	[3.]25	9.1	105583.0
	[3.1],		
26	[3.3]26	9.6	116970.0
	[3.3],		
27	[3.3]27	9.7	112636.0
	[3.8],		
28	[4.]28	10.4	122392.0
	[4.1],		
29	[4.1]29	10.6	121873.0
	[4.2],		
	[4.6],		
	[5.],		
	[5.2],		
	[5.4],		
	[6.],		
	[6.1],		
	[6.9],		
	[7.2],		
	[8.],		
	[8.3],		
	[8.8],		
	[9.1],		
	[9.6],		
	[9.7],		
	[10.4],		
	[10.6]])		

Y

```
array([[ 39344.],
       [ 46206.],
       [ 37732.],
       [ 43526.],
       [ 39892.],
       [ 56643.],
       [ 60151.],
       [ 54446.]])
```

```
[ 64446.],  
[ 57190.],  
[ 63219.],  
[ 55795.],  
[ 56958.],  
[ 57082.],  
[ 61112.],  
[ 67939.],  
[ 66030.],  
[ 83089.],  
[ 81364.],  
[ 93941.],  
[ 91739.],  
[ 98274.],  
[101303.],  
[113813.],  
[109432.],  
[105583.],  
[116970.],  
[112636.],  
[122392.],  
[121873.]])
```

```
from sklearn.model_selection import train_test_split
```

```
X_train,X_test,Y_train,Y_test = train_test_split(  
    X,Y,  
    test_size=0.2,  
    random_state=42  
)
```

X_train

```
array([[10.4],  
       [ 8.8],  
       [ 4.1],  
       [ 1.2],  
       [ 2.3],  
       [ 5.2],
```

```
[ 3. ],  
[ 4.2],  
[ 4.1],  
[ 8. ],  
[ 1.4],  
[ 1.6],  
[ 9.1],  
[ 2.1],  
[ 7.2],  
[ 9.6],  
[ 6. ],  
[10.6],  
[ 6.9],  
[ 3.3],  
[ 4. ],  
[ 4.6],  
[ 6.1],  
[ 3.1]])
```

```
from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
```

```
model.fit(X_train,Y_train)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
Y_pred = model.predict(X_test)
```

```
print("Actual Salary:",Y_test[:5])  
print("Predicted Salary:",Y_pred[:5])
```

```
Actual Salary: [[112636.]  
[ 67939.]
```

```
[113813.]  
[ 83089.]  
[ 64446.]]  
Predicted Salary: [[115791.21011287]  
[ 71499.27809463]  
[102597.86866063]  
[ 75268.80422384]  
[ 55478.79204548]]
```

```
from sklearn.metrics import mean_squared_error  
mse=mean_squared_error(Y_test,Y_pred)  
print("Mean Squared Error:",mse)
```

```
Mean Squared Error: 49830096.855908394
```

```
rmse=np.sqrt(mse)  
print("Root Mean Squared Error:",rmse)
```

```
Root Mean Squared Error: 7059.04362190151
```

```
from sklearn.metrics import r2_score
```

```
R2=r2_score(Y_test,Y_pred)  
print("R2 Score:",R2)
```

```
R2 Score: 0.9024461774180497
```

```
import matplotlib.pyplot as plt
```

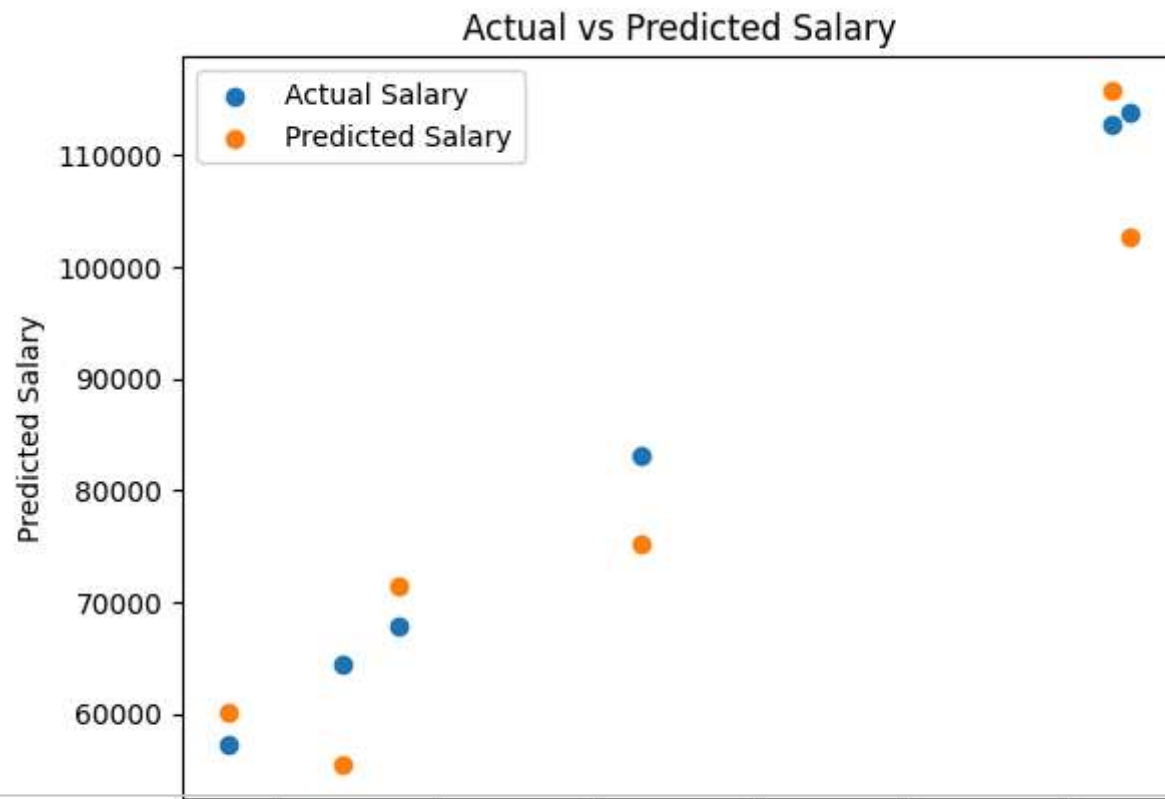
```
plt.scatter(X_test,Y_test,color='Red')  
plt.plot(X_test,Y_pred,color='Blue')  
plt.xlabel('Years of Experience')  
plt.ylabel('Salary')  
plt.title('Salary vsTears of Experience')  
plt.show()
```



```
plt.scatter(Y_test,Y_test,label="Actual Salary")
plt.scatter(Y_test,Y_pred,label="Predicted Salary")

plt.xlabel("Actual Salary")
plt.ylabel("Predicted Salary")

plt.title("Actual vs Predicted Salary")
plt.legend()
plt.show()
```



```
slope=model.coef_  
intercept=model.intercept_  
print("Slope (m):",slope)  
print("Intercept (c):",intercept)
```

```
Slope (m): [[9423.81532303]]  
Intercept (c): [24380.20147947]
```

```
experience=[[5]]
```

```
predicted_salary=model.predict(experience)
```